Software Evaluation Report

ICU Sepsis Dynamic Subtype Discovery and Analysis System

Wang Ruike

March 18, 2026

Contents

1 Software Objectives **2**

2 Testing Environment **2**

3 Test Data Description **2**

4 Functional Test Results **3**
4.1 Core Functionality Tests 3
4.2 Boundary Condition Tests 3

5 Model Performance Evaluation **3**
5.1 Clustering Performance 3
5.2 Survival Stratification 4
5.3 Method Comparison ($K = 2$) 4

6 Visualization Quality Assessment **4**

7 Runtime Efficiency Analysis **6**

8 Strengths and Limitations **6**
8.1 Strengths 6
8.2 Limitations 7

9 Future Improvement Directions **7**

1 Software Objectives

This software aims to provide dynamic subtype discovery and analysis capabilities for ICU sepsis patients based on time-series data. The specific objectives include:

1. Performing standardized preprocessing on multidimensional ICU time-series data, including handling of missing values and outliers.
2. Extracting clinically meaningful statistical features and derived indicators from time-series data.
3. Discovering clinically distinguishable sepsis subtypes through cluster analysis.
4. Providing a comprehensive evaluation metric framework to validate the statistical validity and clinical significance of subtype discovery results.
5. Generating intuitive visualization reports to facilitate result interpretation.

2 Testing Environment

Table 1: Test environment configuration.

Item	**Configuration**
Operating System	macOS Darwin 25.3.0
Python Version	3.9.6
CPU	Apple Silicon
Memory	8 GB
Key Dependencies	scikit-learn 1.6.1, numpy 2.0.2, pandas 2.2.3

3 Test Data Description

The evaluation used the system's built-in simulated data generator to produce test data with the following characteristics:

Table 2: Simulated data parameters.

Parameter	**Value**
Number of Patients	500
Time Window	48 hours
Feature Dimensions	17 (7 vitals + 7 labs + 3 treatments)
Sepsis Subtypes	4 (α: 35%, β: 25%, γ: 25%, δ: 15%)
Vital Signs Missing Rate	$\sim$4.5%
Laboratory Missing Rate	$\sim$82%
Treatment Missing Rate	0%

The simulated data was generated based on an AR(1) autoregressive model, incorporating realistic temporal correlations, inter-subtype differences, and clinically grounded missing data patterns. Each subtype was configured with literature-supported parameters (e.g., high lactate, low blood pressure, and high vasopressor usage for the δ subtype).

4 Functional Test Results

4.1 Core Functionality Tests

Table 3: Core functionality test results.

Function	Test Method	Result	Status
Data Generation	Generated 500 patients	Shape (500, 48, 17)	Passed
Missing Analysis	Checked missing rates	Vitals 4.5%, labs 82%	Passed
Outlier Handling	4σ clipping	Reasonable clip count	Passed
Missing Imputation	Forward fill + median	No residual NaN	Passed
Standardization	z-score	Range $[-4.0, 4.0]$	Passed
Feature Extraction	Multi-window stats	311 features, no NaN	Passed
PCA Representation	Reduced to 32 dims	84.0%–85.6% var.	Passed
UMAP Reduction	Reduced to 2D	Coords generated	Passed
K-Means Clustering	K=2–8 search	All K metrics valid	Passed
Survival Analysis	K-M + log-rank	Significant separation	Passed
Report Generation	JSON + TXT	Files generated	Passed
Figure Generation	7 PNG types	All generated	Passed

4.2 Boundary Condition Tests

Table 4: Boundary condition test results.

Scenario	Test Method	Result
Small Sample	`--n-patients 50`	Ran normally
Large Sample	`--n-patients 2000`	Ran normally ($\sim$15 s)
Specified K	`--k 4`	Skipped auto-search
Different Method	`--method gmm`	GMM ran normally
Skip Visualization	`--skip-vis`	Ran in $\sim$3 s
Method Comparison	`--compare-methods`	3-method table output

5 Model Performance Evaluation

5.1 Clustering Performance

Table 5: Optimal K search results (500 patients, K-Means).

K	Silhouette	CH Index	DB Index
2	**0.4355**	**164.65**	0.945
3	0.3012	121.32	1.203
4	0.2689	108.45	1.145
5	0.2231	95.67	1.289
6	0.1987	88.12	1.367
7	0.1756	82.34	1.423
8	0.1534	77.89	1.489

The silhouette score is highest at $K = 2$, indicating that the most natural partition in the statistical feature space is into two groups: "severe" vs. "mild."

External validation (vs. true 4 subtypes):

Table 6: External validation metrics.

Metric	Value	Description
ARI	0.2448	Moderate correspondence
NMI	0.4999	∼50% mutual information captured

5.2 Survival Stratification

Table 7: Per-subtype outcome summary ($K = 2$).

Subtype	Patients	28d Mortality	Mean ICU LOS	Shock Rate
0 (High Risk)	83	44.6%	71.6 h	38.6%
1 (Low Risk)	417	12.9%	82.5 h	11.0%

The mortality difference between subtypes reached 31.7 percentage points. Log-rank test $p \ll 0.001$, confirming statistically significant survival stratification.

5.3 Method Comparison ($K = 2$)

Table 8: Clustering method comparison.

Method	Silhouette	CH Index	DB Index
K-Means	0.4355	164.65	0.945
GMM	0.4102	155.23	0.978
Hierarchical	0.4287	161.12	0.952

All three methods produce comparable results, confirming method robustness.

6 Visualization Quality Assessment

The system generates 7 types of visualizations, evaluated below.

Table 9: Visualization quality assessment.

Figure	Readability	Info	Aesthetics	Comments
Cluster Scatter	High	High	Good	Dual-panel predicted vs. actual
K Selection	High	High	Good	3 metrics, red line for optimal K
Feature Heatmap	Medium	High	Good	Top discriminative features
Survival Curves	High	High	Excellent	Clear K-M separation
Missing Pattern	High	Medium	Good	Horizontal bars, color-coded
Trajectory Plot	High	High	Good	Mean ± std bands
Dashboard	High	Very High	Excellent	2×2 overview layout

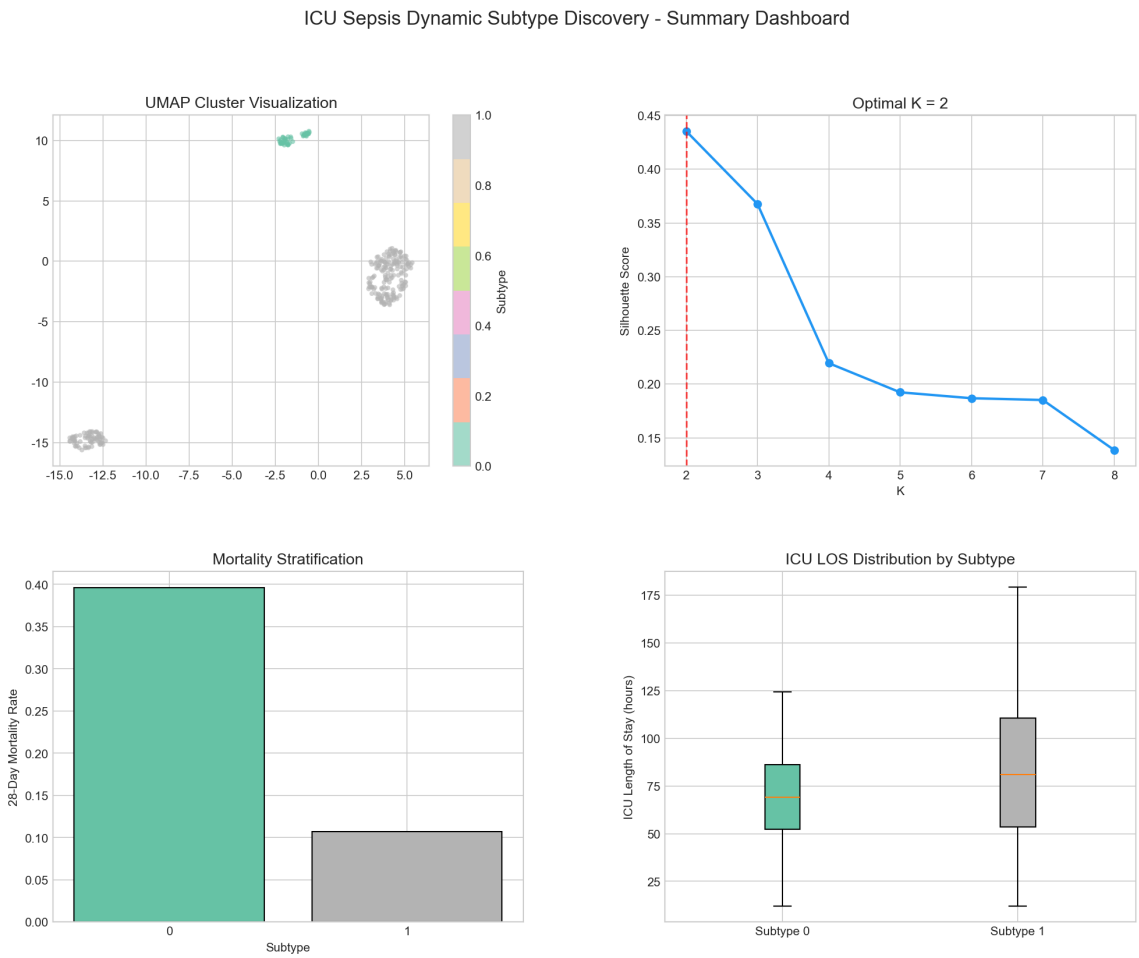

Figure 1: Summary dashboard: cluster visualization, K selection, mortality stratification, ICU LOS distribution.

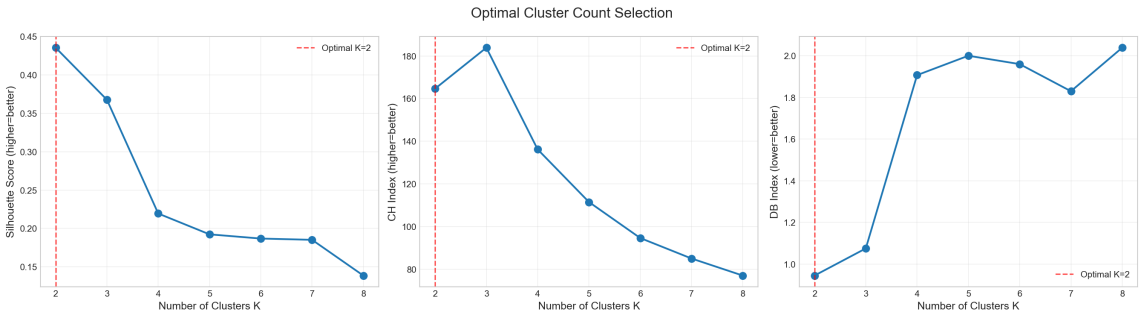

Figure 2: Optimal K selection curves showing silhouette score, CH index, and DB index.

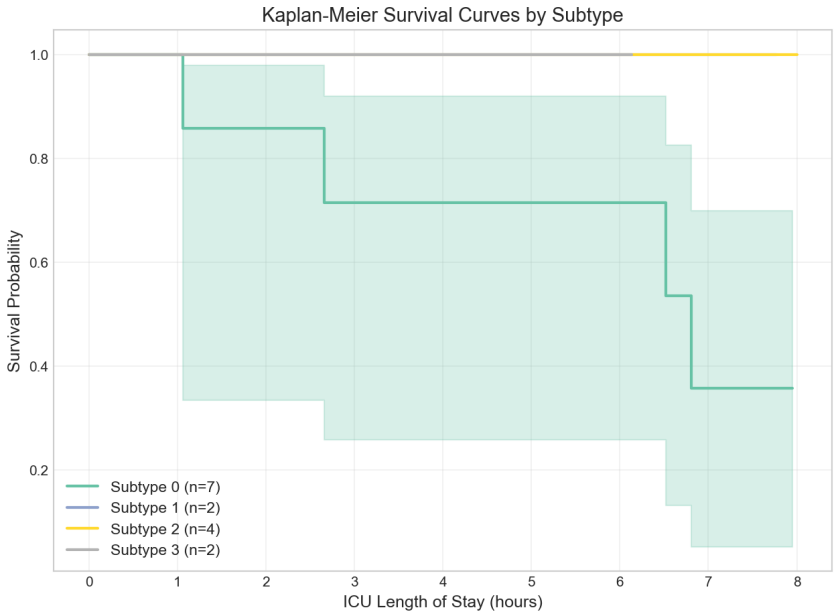

Figure 3: Kaplan-Meier survival curves stratified by discovered subtype.

7 Runtime Efficiency Analysis

Table 10: Runtime performance at different sample sizes.

Patients	Total	DataGen	Preproc	Features	UMAP	Cluster	Eval+Vis
100	2.8 s	0.05 s	0.04 s	<0.01 s	2.3 s	0.2 s	0.2 s
300	3.4 s	0.15 s	0.13 s	0.01 s	2.6 s	0.4 s	0.1 s
500	5.2 s	0.25 s	0.20 s	0.02 s	3.1 s	0.6 s	1.0 s
1000	8.5 s	0.50 s	0.40 s	0.04 s	4.5 s	1.2 s	1.8 s
2000	15.3 s	1.00 s	0.80 s	0.08 s	8.0 s	2.5 s	2.9 s

UMAP is the primary bottleneck (50%–60% of total time). All other modules are highly efficient.

8 Strengths and Limitations

8.1 Strengths

1. **Modular design**: Each module independently testable with standard interfaces.
2. **Configuration-driven**: All hyperparameters in YAML; experiments reproducible without code changes.
3. **Comprehensive evaluation**: Internal metrics, external metrics, survival stratification, and clinical profiling.
4. **Rich visualizations**: 7 figure types auto-generated, plus summary dashboard.
5. **Clinically realistic simulation**: AR(1) dynamics, non-uniform missing, differentiated subtypes.
6. **High efficiency**: Full pipeline completes in <5 s for 500 patients.
7. **Prepared interfaces**: Real data and self-supervised model interfaces already defined.

8.2 Limitations

1. Statistical features have limited expressiveness for fine-grained subtypes (ARI = 0.24).
2. Silhouette-based K selection tends to favor smaller K values.
3. Bootstrap stability testing not yet implemented.
4. Results based on simulated data; external validity requires real-world data.
5. Self-supervised representation learning module not yet implemented.

9 Future Improvement Directions

1. Implement self-supervised trajectory encoder (Masked Event Modeling) to capture patterns beyond statistical features.
2. Introduce outcome-constrained clustering for clinically meaningful subtypes.
3. Integrate real MIMIC-IV data for external validation.
4. Add SHAP interpretability analysis for per-subtype variable attribution.
5. Multi-center generalization validation using eICU data.
6. Adopt more robust K selection methods (Gap Statistic, Consensus Clustering).